

House Price Prediction Project

1. Project Overview and Objectives

Project Overview

This project predicts customer churn using features like tenure, monthly charges, and contract type. It includes data preprocessing, feature engineering, and model training with classification algorithms. Key factors driving churn are identified to support proactive retention strategies.

Objectives

- To analyze customer data to understand factors influencing churn.
- To identify key features affecting churn (tenure, charges, contract, payment).
- To engineer and preprocess features for effective machine learning modeling.
- To build and evaluate predictive models for accurate churn prediction.
- To generate actionable insights for improving customer retention.

2. Setup and Installation Instructions

Prerequisites

- Python 3.10+
- Jupyter Notebook

Installation Steps

1. Download or clone the project repository from GitHub.
2. Install required libraries:

```
pip install -r requirements.txt
```

3. Place dataset inside /data folder.
4. Launch Jupyter Notebook and open the notebook:

```
notebook/churn_prediction_pipeline.ipynb
```

3. Code Structure Explanation

```
Customer_churn_prediction_week10/
|
├── data/
|   └── customer_churn.csv
├── notebooks/
|   └── churn_prediction_pipeline.ipynb
├── src/
|   ├── data_loader.py
|   ├── preprocessing.py
|   ├── feature_engineering.py
|   ├── outlier_handler.py
|   └── pipeline_builder.py
├── reports/
|   ├── preprocessing_report.md
|   └── feature_engineering_documentation.md
├── screenshots/
|   └── Images.png
|
├── model_bundle.pkl
├── requirements.txt
└── README.md
```

File Descriptions

- data/customer_churn.csv
Dataset containing customer information and churn labels (0 = retained, 1 = churned).
- notebooks/churn_prediction_pipeline.ipynb
Jupyter notebook implementing the full workflow: data exploration, preprocessing, feature engineering, model building, and evaluation.

- `src/data_loader.py`
Functions to load and inspect the dataset.
- `src/preprocessing.py`
Functions for handling missing values, encoding, scaling, and pipelines.
- `src/feature_engineering.py`
Functions for creating new features to improve model performance.
- `src/outlier_handler.py`
Functions for detecting and handling outliers using IQR and Z-score methods.
- `src/pipeline_builder.py`
Builds the complete ML pipeline with preprocessing and model.
- `screenshots/`
Contains visualizations such as feature importance, churn distribution, and model confusion matrices.
- `report/preprocessing_report.md`
Documentation of all preprocessing steps and outcomes.
- `report/feature_engineering_documentation.md`
Details of engineered features, their motivation, and expected impact on model performance.
- `Model_bundle.pkl`
Serialized machine learning pipeline with trained model.
- `requirements.txt`
Specifies all Python dependencies needed to run the project.
- `README.md`
Project documentation including overview, setup instructions, and usage guide.

4. Screenshots of Working Application

The included screenshots demonstrate:

- The Churn Distribution plot shows most customers do not churn.
- The Monthly Charges vs Churn plot shows churners have higher monthly charges.
- Feature importance identifies Tenure as the strongest predictor, followed by TotalCharges and MonthlyCharges.
- The Confusion Matrix shows high accuracy and perfect recall for churners.
- Model evaluation results (Random Forest Accuracy: 0.97, ROC-AUC: 0.99) confirm strong predictive performance.

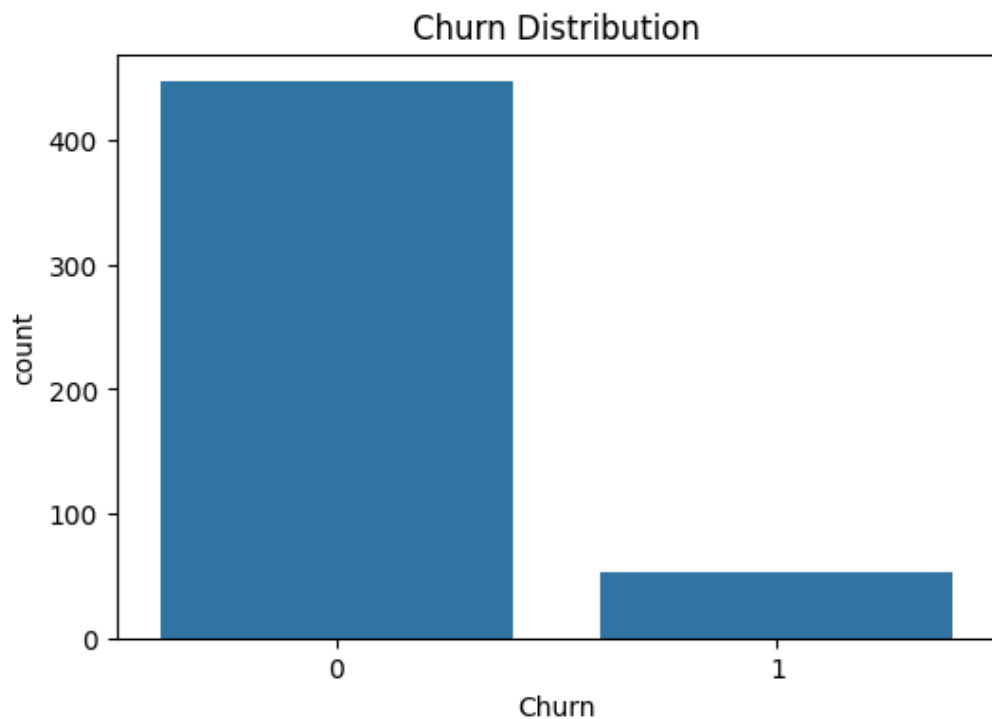


Fig. 4.1. Churn Distribution Plot

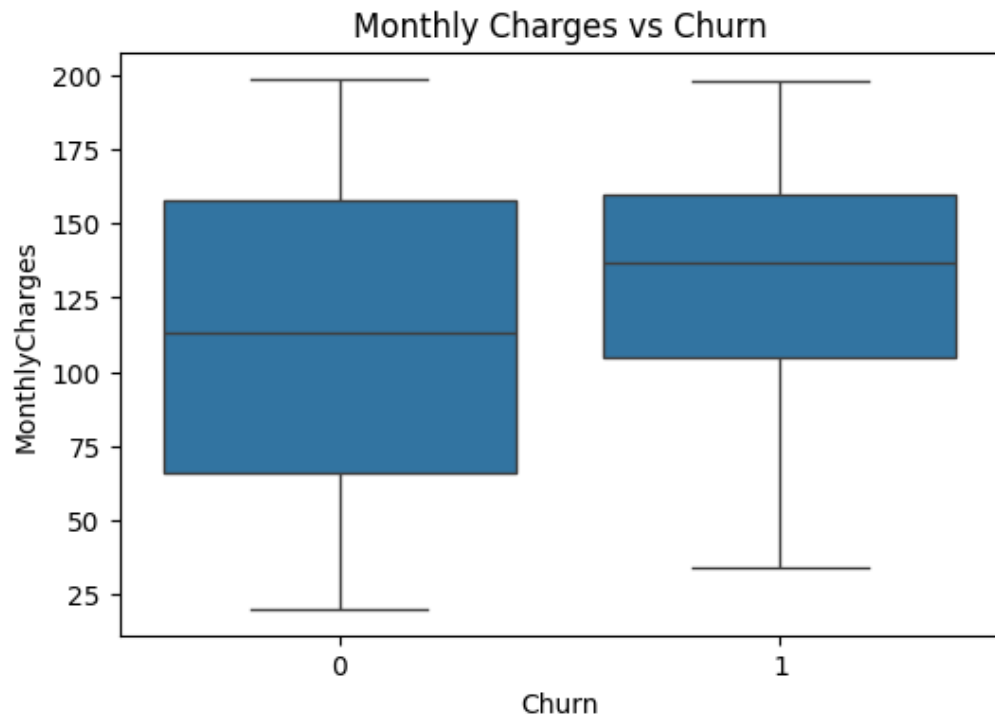


Fig. 4.2. Monthly Charges vs Churn Boxplot

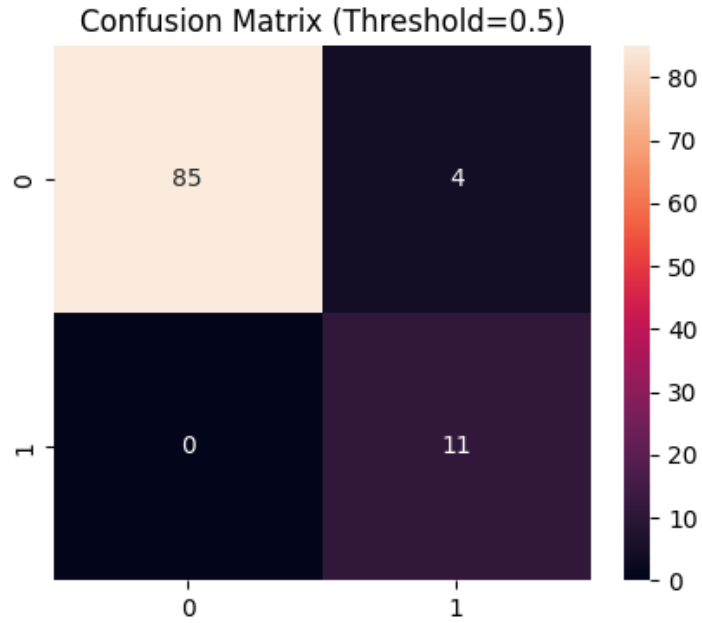


Fig. 4.3. Confusion Matrix – Subset / Validation

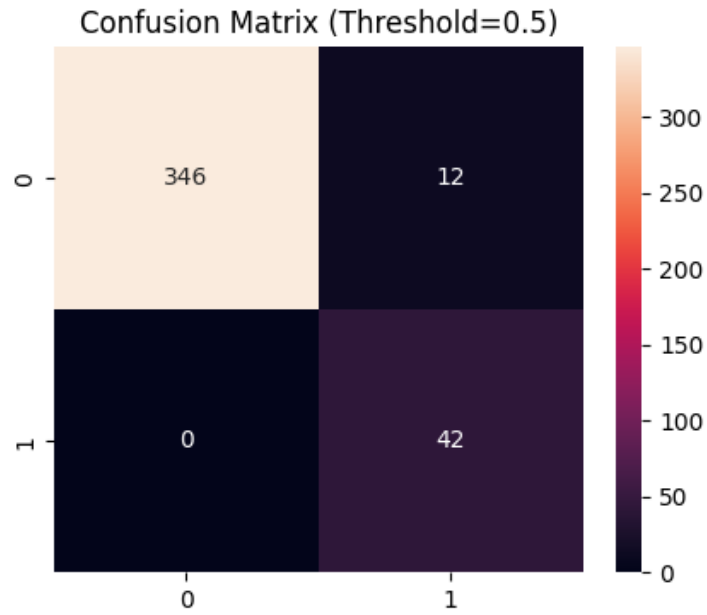


Fig. 4.4. Confusion Matrix – Full Dataset

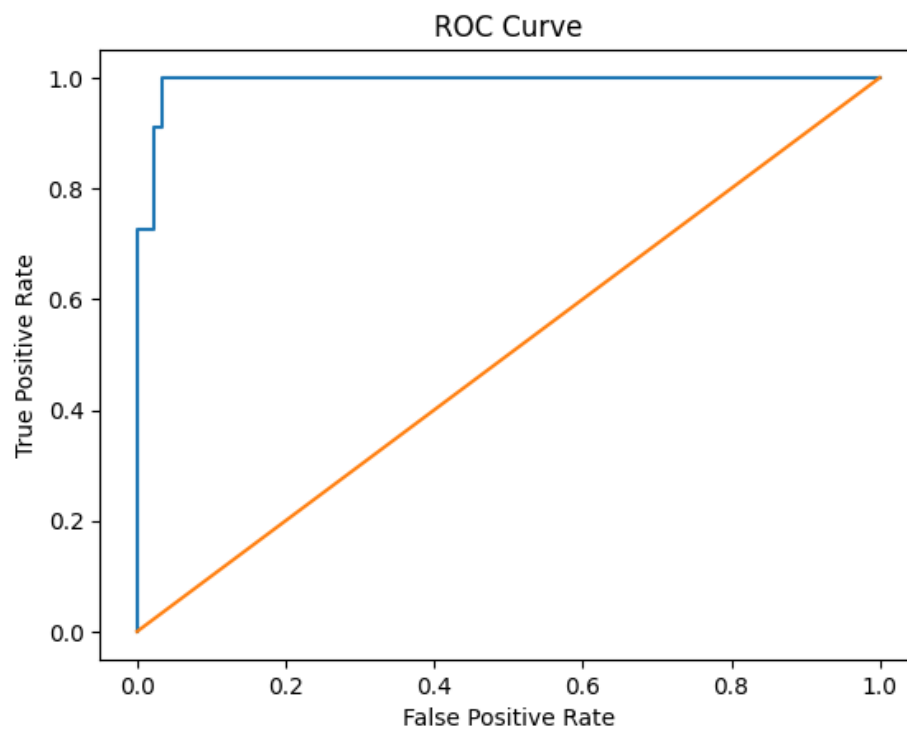


Fig. 4.5. ROC Curve – Model Performance

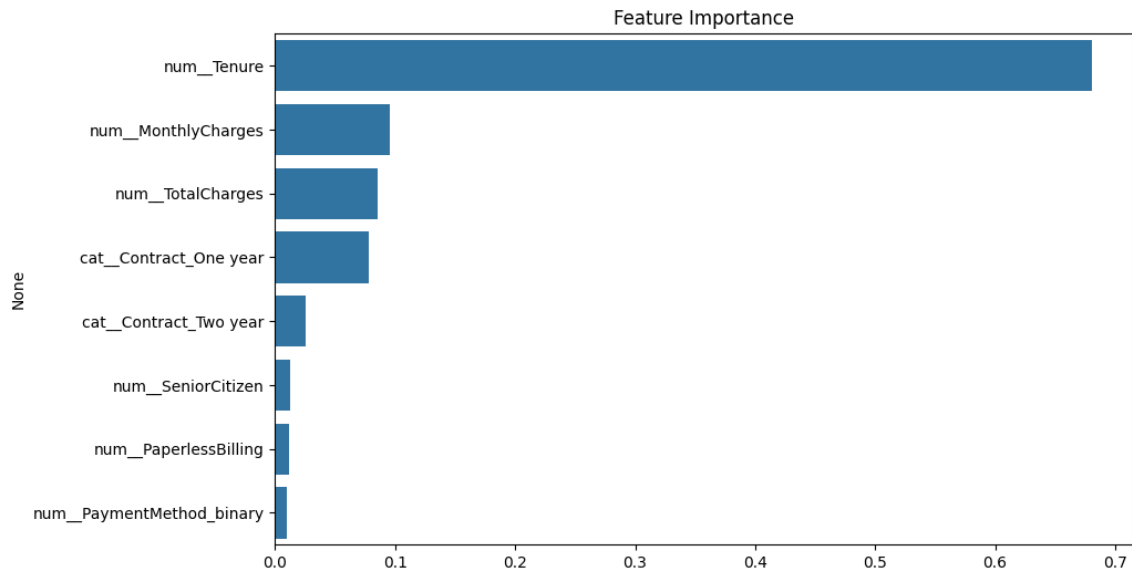


Fig. 4.6. Feature Importance Plot

These results validate that the data preprocessing, feature engineering, and machine learning pipeline were successfully implemented, providing clear insights into key factors influencing customer churn.

5. Technical Requirements Fulfillment Explanation

Category	Technical Requirement	How It Was Implemented
Algorithm	Data loading and preprocessing	Churn dataset loaded using Pandas, checked for missing values, and categorical variables identified for encoding.
Algorithm	Data exploration	.info(), .describe(), and .isnull() used to understand data types, distribution, and completeness; Churn distribution plotted with Seaborn.

Category	Technical Requirement	How It Was Implemented
Algorithm	Handling categorical data	Label Encoding applied to PaperlessBilling; One-Hot Encoding applied to Contract; Binary encoding applied to PaymentMethod.
Algorithm	Feature scaling	StandardScaler and MinMaxScaler applied to Tenure, MonthlyCharges, and TotalCharges; compared model performance after scaling.
Algorithm	Outlier detection	Interquartile Range (IQR) and Z-score methods applied; no significant outliers found.
Algorithm	Feature engineering	Created new features: CustomerLifetimeValue, AvgRevenuePerMonth, PaymentEfficiency, SeniorCitizenContract, ChargeDifference.
Algorithm	Feature selection	Random Forest feature importance calculated to identify most impactful features for churn prediction.
Algorithm	Predictive modeling	RandomForestClassifier implemented with Scikit-learn; cross-validation used to evaluate performance.
Algorithm	Model optimization	GridSearchCV used to tune hyperparameters: max_depth, min_samples_leaf, min_samples_split, and n_estimators.

Category	Technical Requirement	How It Was Implemented
Algorithm	Model evaluation	Confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC metrics calculated for train/test splits.
Data Structure	Tabular data handling	Pandas DataFrames used to store and transform structured customer data.
Data Structure	Feature storage	Encoded categorical features and scaled numerical features stored as arrays for ML input.
Architecture	Modular project design	Code organized into separate modules: data_loader.py, preprocessing.py, feature_engineering.py, pipeline_builder.py, and notebooks.
Architecture	Reproducible ML pipeline	Complete preprocessing, feature engineering, and model training executed through Scikit-learn pipelines for reproducibility.
Output	Visualization outputs	Visualizations include Churn distribution, MonthlyCharges vs Churn boxplot, Feature Importance chart, and Confusion Matrix.
Output	Model evaluation metrics	Model performance evaluated using Accuracy, ROC-AUC, Precision, Recall, and F1-score.

Category	Technical Requirement	How It Was Implemented
Output	Model comparison insights	StandardScaler and MinMaxScaler achieved identical accuracy (0.98); final RandomForestClassifier model achieved 0.97 accuracy and 0.99 ROC-AUC.

6. Learning Outcomes

Through this project, the following learning outcomes were achieved:

- Gained hands-on experience in building an end-to-end machine learning pipeline in Python.
- Learned data preprocessing techniques including categorical encoding, scaling, outlier handling, and feature engineering.
- Developed skills in exploratory data analysis and visualization to understand customer behavior and churn patterns.
- Built and evaluated models using Random Forest, optimizing performance through hyperparameter tuning.
- Applied model evaluation metrics such as accuracy, ROC-AUC, precision, recall, and F1-score.
- Understood feature importance to identify key drivers of customer churn.

7. Conclusion

This project demonstrated how machine learning can predict customer churn using telecom data. Through data exploration, preprocessing, and feature engineering, key factors influencing churn—such as tenure and monthly charges—were identified. A Random Forest model achieved high accuracy and ROC-AUC, showing that data-driven approaches can help businesses proactively retain at-risk customers.